EECE 306 Lab 9 notebook. Current Sources and the Biot-Savart Law

Team 03 — Carissa McWilliams, Amelia Harris, and Mitch Nazareth.

Contents

- Setup
- How long em.field.H took
- Circular loop against the closed form
- Finite segment against the infinite wire
- Solenoid, built by reusing the ring
- Reversal, linearity, and the merge policy
- Timing at N = 5000
- Interpretation
- Problems encountered
- Full test suite

Setup

```
clear; close all;
```

How long em.field.H took

em.field.H is the same loop as em.field.E with a different numerator.

```
minutes_for_H = 70;
% The biggest challenge was taking the structure from em.field.E and
% changing it to calculate the magnetic field using the Biot-Savart law.
% I had to make sure the cross product was handled correctly and that
% the current source elements and their tangent vectors were being used
% properly. I also had to make sure the function worked with multiple
% observation points and handled points where the field calculation
% could become singular. Reusing the source and integration code from
% the previous labs helped save a lot of time. I think this would have
% gone faster if the current source structure and tangent vector
% calculations had already been set up.
```

Circular loop against the closed form

```
a = 1; I = 1;
ring = @(t)[a*cos(t) a*sin(t) zeros(size(t))];
sr = em.src.lineCurrent(I, ring, [0 2*pi], 400);
z = linspace(0, 2, 10)';
Hnum = em.field.H(sr, [zeros(10,2) z]);
Href = I*a^2 ./ (2*(a^2 + z.^2).^1.5);
disp('    z (m)     relative error')
disp([z abs(Hnum(:,3) - Href)./Href])
Hc = em.field.H(sr, [0 0 0]);
fprintf('center H_z = %.6f   I over 2a = %.6f\n', Hc(3), I/(2*a));
fprintf('transverse component at center = %.3e  (consistent with zero)\n', hypot(Hc(1), Hc(2)));
```

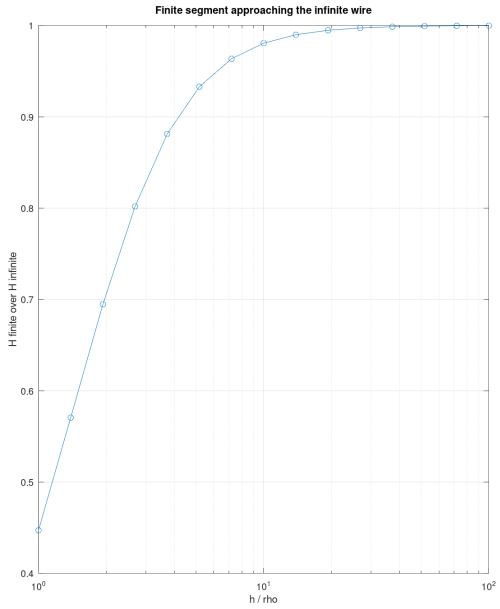
```
       z (m)     relative error
            0   0.0000
       0.2222   0.0000
       0.4444   0.0000
       0.6667   0.0000
       0.8889   0.0000
       1.1111   0.0000
       1.3333   0.0000
       1.5556   0.0000
       1.7778   0.0000
       2.0000   0.0000
   center H_z = 0.500000   I over 2a = 0.500000
   transverse component at center = 0.000e+00  (consistent with zero)
```

Finite segment against the infinite wire

The same study as the Lab 4 finite line, with the field engine swapped. The curve has the same shape, and the handout asks why.

```
rho = 1;
hor = logspace(0, 2, 15)';
ratio = zeros(size(hor));
for k = 1:numel(hor)
    h = hor(k) * rho;
    seg = @(t)[zeros(size(t)) zeros(size(t)) t];
    ss = em.src.lineCurrent(I, seg, [-h/2 h/2], 800);
    Hs = em.field.H(ss, [rho 0 0]);
    ratio(k) = abs(Hs(2)) / (I/(2*pi*rho));
end
figure; semilogx(hor, ratio, 'o-'); grid on
xlabel('h / rho'); ylabel('H finite over H infinite')
title('finite segment approaching the infinite wire')
fprintf('ratio first exceeds 0.99 at h/rho about %.1f\n', hor(find(ratio > 0.99, 1)));
```

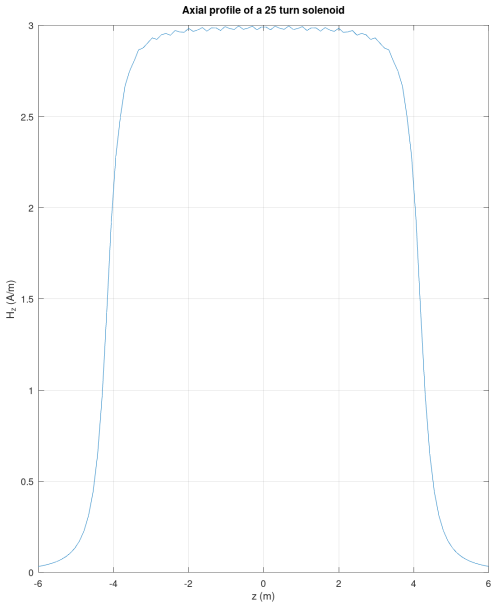
```
   ratio first exceeds 0.99 at h/rho about 19.3
```



Solenoid, built by reusing the ring

```
aS = 0.4; hS = 8; Nturns = 25;
zk = linspace(-hS/2, hS/2, Nturns);
srcs = cell(1, Nturns);
for k = 1:Nturns
    zc = zk(k);
    srcs{k} = em.src.lineCurrent(I, @(t)[aS*cos(t) aS*sin(t) zc*ones(size(t))], [0 2*pi], 200);
end
sol = em.src.merge(srcs{:});
zax = linspace(-6, 6, 100)';
Hax = em.field.H(sol, [zeros(100,2) zax]);
figure; plot(zax, Hax(:,3), '-'); grid on
xlabel('z (m)'); ylabel('H_z (A/m)')
title('Axial profile of a 25 turn solenoid')
Hcen = em.field.H(sol, [0 0 0]);
fprintf('center H_z = %.4f   N I over h = %.4f   ratio = %.4f\n', ...
    Hcen(3), Nturns*I/hS, Hcen(3)/(Nturns*I/hS));
fprintf('aspect ratio h/a = %.0f\n', hS/aS);
```

```
   center H_z = 2.9978   N I over h = 3.1250   ratio = 0.9593
   aspect ratio h/a = 20
```



Reversal, linearity, and the merge policy

```
sneg = em.src.lineCurrent(-I, ring, [0 2*pi], 400);
fprintf('max |H(-I) + H(I)|  = %.3e  (reversal)\n', ...
    max(em.vec.mag(em.field.H(sneg, [0.3 0.2 0.5]) + em.field.H(sr, [0.3 0.2 0.5]))));
s2I = em.src.lineCurrent(2*I, ring, [0 2*pi], 400);
fprintf('max |H(2I) - 2H(I)| = %.3e  (linearity)\n', ...
    max(em.vec.mag(em.field.H(s2I, [0.3 0.2 0.5]) - 2*em.field.H(sr, [0.3 0.2 0.5]))));
sm = em.src.merge(sr, s2I);
fprintf('merged current source element count = %d  (sum of parts = %d)\n', ...
    size(sm.pos,1), size(sr.pos,1) + size(s2I.pos,1));
try
    em.src.merge(sr, em.src.pointCharge(1e-9, [0 0 0]));
```

```
    disp('ERROR, this line must not be reached')
catch err
    fprintf('merging current with charge correctly threw. %s\n', err.message);
end
```

```
   max |H(-I) + H(I)|  = 0.000e+00  (reversal)
   max |H(2I) - 2H(I)| = 0.000e+00  (linearity)
   merged current source element count = 800  (sum of parts = 800)
   merging current with charge correctly threw. Cannot merge a current source with a charge source.
```

Timing at N = 5000

```
pts = randn(5000,3) + 5;
tic; em.field.H(sr, pts); t = toc;
fprintf('em.field.H on 5000 points took %.3f s  (requirement, under 2 s)\n', t);
```

```
   em.field.H on 5000 points took 0.621 s  (requirement, under 2 s)
```

Interpretation

Lab 3 made this lab easier because the field engine was already set up to calculate fields at multiple observation points. Lab 4 also helped since I could reuse the source structures and numerical integration methods instead of starting over. Most of the work for em.field.H was changing the calculation to use the Biot-Savart law rather than having to build a completely new field engine. Going forward, I would try to design interfaces so they can be reused for different types of sources and field calculations. This would make adding new features easier and avoid having to rewrite code that already works.

Problems encountered

None, this lab went smoothly.

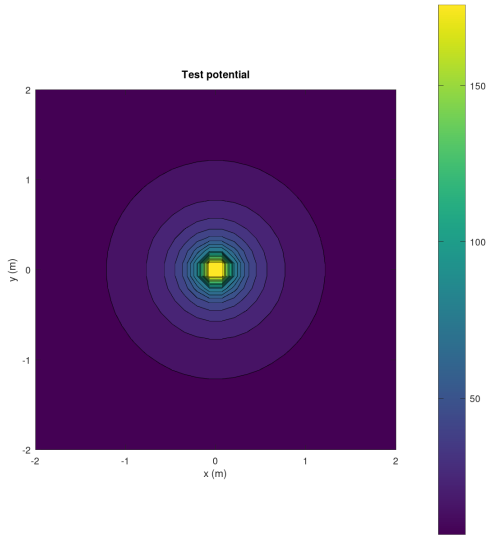
Full test suite

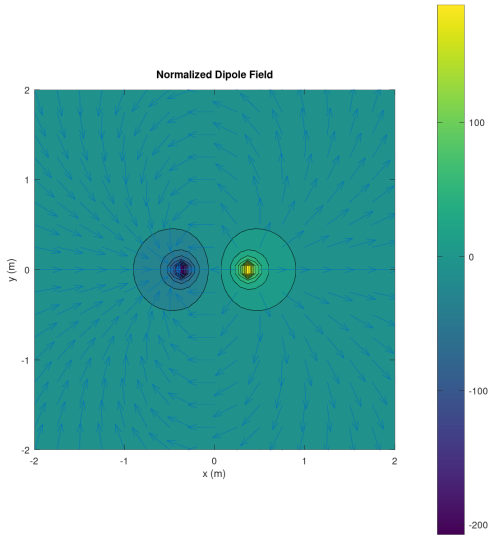
```
runTests
```

```
   EECE 306 toolbox test suite
   ---------------------------------------------------------------------
   11.3.0
   eps0 = 8.8541878128e-12 F/m
   mu0  = 1.2566370614e-06 H/m
   c0   = 2.99792458e+08 m/s
   eta0 = 376.730314 ohm
   Derived constant checks passed.
   mag([3 4 0])  = 5   (expect 5 exactly)
   unit([3 4 0]) =
      0.6000   0.8000        0
   unit([0 0 0]) (expect all NaN) =
      NaN   NaN   NaN
   angle perpendicular = 1.5707963267949  (expect pi/2)
   angle antiparallel  = 3.14159265358979, real = 1  (expect pi, 1)
   fromTo([1 1 1],[2 3 4]) =
      1     2     3
   mag    input 100x3 output [100 1]
   unit   input 100x3 output [100 3]
   angle  input 100x3 output [100 1]
   fromTo input 100x3 output [100 3]
     PASS  test_lab01            0.01 s
   spherical   roundtrip max error = 5.947e-15
   cylindrical roundtrip max error = 2.047e-15
   c2sph([0 0 1])  =
      1     0     0
   c2sph([0 0 -1]) =
      1.0000   3.1416        0
   phi in the four quadrants = 0.7854 2.3562 3.9270 5.4978 rad
   all inside [0, 2*pi) = 1
   vector roundtrip (spherical pair) max error = 8.882e-16
   magnitude change under conversion max       = 8.882e-16
   vector roundtrip (cylindrical pair) max error = 6.280e-16
   at (0, 5, 0), (Ar, Atheta, Aphi) =
      3.0000e+00   1.8370e-16  -2.0000e+00
   at (4, 0, 0), (Ar, Atheta, Aphi) =
     2.0000e+00   1.2246e-16   3.0000e+00
     PASS  test_lab02            0.02 s
      r (m)    relative error
       1.0000         0
       2.0000         0
       3.0000         0
       4.0000         0
       5.0000         0
       6.0000         0
       7.0000         0
       8.0000         0
       9.0000    0.0000
      10.0000         0
   max |Ex| on bisector      = 0.000e+00 V/m
   max |E| on the same plane = 6.298e+02 V/m
   |E(r)| / |E(2r)| = 8.0003   (expect near 8 for a dipole)
   em.field.E on 5000 points took 0.001 s  (requirement, under 2 s)
     PASS  test_lab03            0.28 s
     test_lab04 checks complete
     PASS  test_lab04            0.33 s
   div D = 1.003980e-06, rho_v = 1.000000e-06
     test_lab05 checks complete
     PASS  test_lab05            0.37 s
   warning: The represented source extent is at least 50 times the median nearest-element distance for these observation points. Check finite extent; V = 0 at infinity is not valid for an ideal unbounded source.
   warning: called from
       V at line 51 column 9
       test_lab06 at line 232 column 1
       run at line 78 column 7
       runTests at line 73 column 9
       publish>eval_code_helper at line 1081 column 3
       publish>eval_code at line 997 column 11
       publish at line 402 column 5

   warning: The represented source extent is at least 50 times the median nearest-element distance for these observation points. Check finite extent; V = 0 at infinity is not valid for an ideal unbounded source.
   warning: called from
       V at line 51 column 9
       test_lab06 at line 235 column 1
       run at line 78 column 7
       runTests at line 73 column 9
       publish>eval_code_helper at line 1081 column 3
       publish>eval_code at line 997 column 11
       publish at line 402 column 5

     test_lab06 checks complete
     PASS  test_lab06            1.57 s
     Lab 7 MMS error 8.224e-05; observed order 2.055; Gauss-law relative error 0.000
     test_lab07 checks complete
     PASS  test_lab07           73.75 s
     Lab 8 parallel-plate rel err 1.82e-16; two-layer harmonic order unresolved (aligned 1-D case); arithmetic order 0.983; coax rel err 0.0149
     test_lab08 checks complete
     PASS  test_lab08            4.62 s
     Lab 9 loop center Hz 2.875 A/m; wire ratio 0.99228 at h/rho=8.0; solenoid ratio 0.99123 at h/a=15.0; 5000-point H 0.024 s
     test_lab09 checks complete
     PASS  test_lab09            0.11 s
   ---------------------------------------------------------------------
     9 passed, 0 failed, 9 total
     All tests passing.
```





Published with GNU Octave 11.3.0